A MICROFLUIDC SYSTEM FOR LABEL-FREE AND BIO-COMPATIBLE MAGNETIC SEPARATION OF SMALL EXTRACELLULAR VESICLES

Lin Zeng[1], Shi Hu[1], Xi Chen[1], Pengcheng Zhang[1], Guoqiang Gu[1], Yuye Wang[1], Hongpeng Zhang[2], and Hui Yang[1,*]

[1]*Laboratory of Biomedical Microsystems and Nano Devices, Bionic Sensing and Intelligence Center, Institute of Biomedical and Health Engineering, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, CHINA*

[2]*Marine Engineering College, Dalian Maritime University, CHINA*

ABSTRACT

A microfluidic system for label-free and biocompatible magnetic separation of small extracellular vesicles is reported. By using the on-chip magnetic pole arrays, the magnetic field gradient in the microchannel is drastically increased. Therefore, the on-chip negative magnetophoretic separation with nanoscale resolution is realized using the ferrofluid of extremely low concentration. In the experiments, the sEVs are successfully separated from cell culture supernatant, the recovery rate and purity of sEVs are 85.80% and 80.45%, superiorly exceeding the performance that ultracentrifugation method can provide.

KEYWORDS: Microfluidic system, Magnetic separation, Negative magnetophoresis, Small extracellular vesicles

INTRODUCTION

As the carriers of biomolecules to transfer between cells, small extracellular vesicles (sEVs) are related to many physiological and pathological processes [1]. In order to conduct in-depth analysis and utilize them for further applications, sEVs should be extracted from complex samples first. However, efficient separation of sEVs is difficult due to their small size and low buoyancy density. Negative magnetophoresis shows superior performance in separation of microspheres without labeling [2]. For nanoscale bio-samples, resolution on sample size and biocompatibility of working medium have been the two key issues hindering the application of negative magnetophoresis [3,4]. The proposed system solves the problems and realize efficient separation of sEVs [5].

THEORY

The microfluidic system (Figure 1 A) consists of a PDMS microfluidic chip and a magnetic field module. The magnetic field module incorporates four NdFeB magnets (N52), two pieces of high permeability alloy (permalloy) and two on-chip magnetic pole arrays. The simulation results (Figure 1 B) show that the system achieves the separation of 200 nm and 800 nm particles.

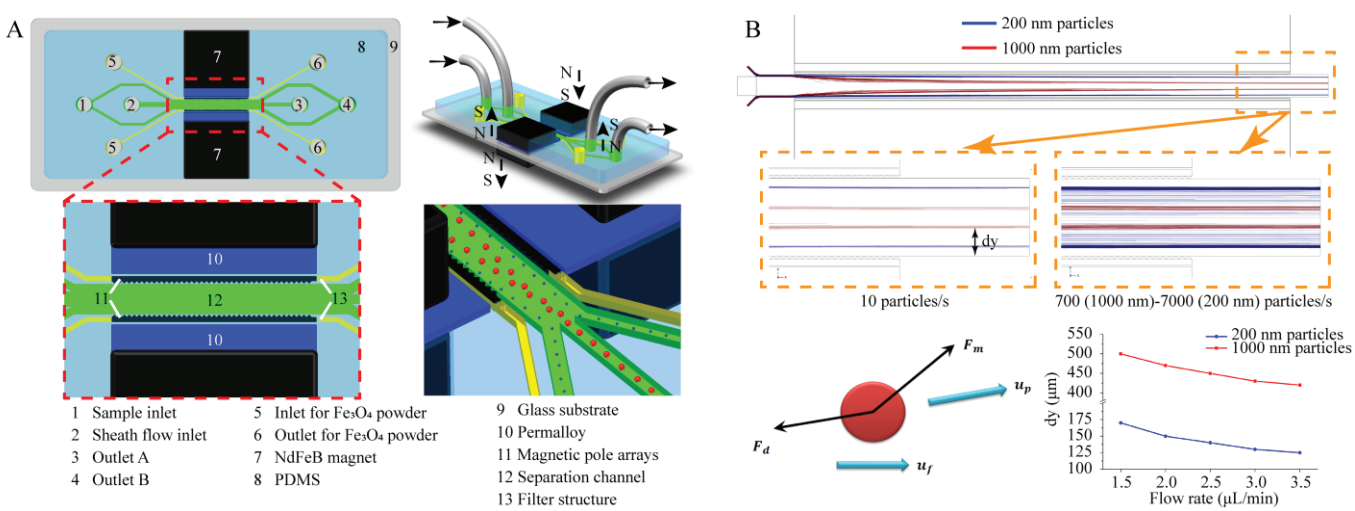

Figure 1: A) Schematic illustration of the magnetic separation system; B) The force analysis and particle trajectories of nonmagnetic particles in the separation channel.

RESULTS AND DISCUSSION

The sEVs in Human Fetal Bone Marrow Mesenchymal Stem Cells (BMSCs) culture supernatant is separated using the system, and collected from the outlet B. The separation results are compared with those obtained from the ultracentrifugation method, as shown in Figure 2 A. The recovery rate and purity of sEVs collected from outlet B are 85.80% and 80.45%, respectively. Furthermore, the membrane structure of the sEVs is not damaged, the expression of sEV membrane tetraspanin is similar to the sEVs extracted by ultracentrifugation.

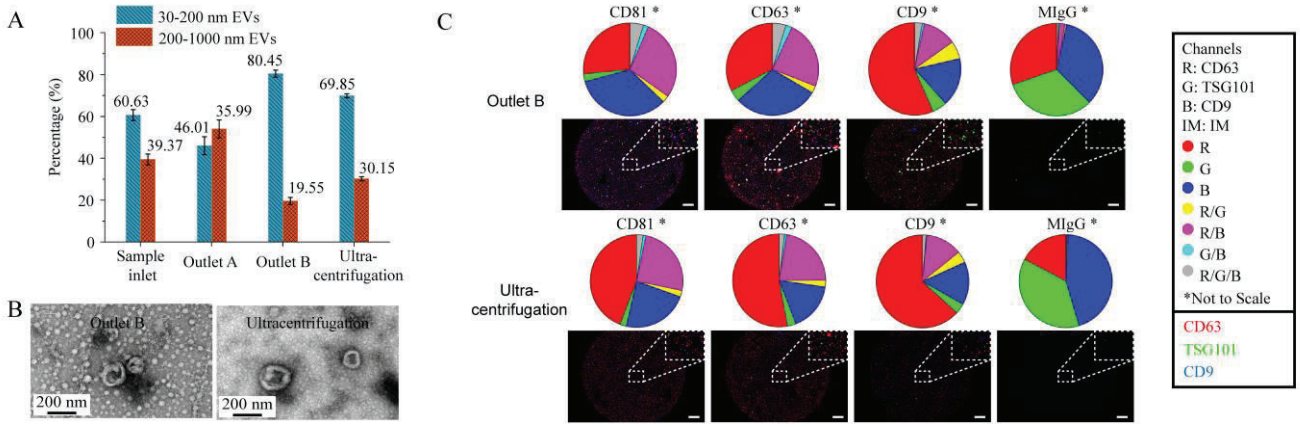

Figure 2: A) Comparison on the percentages of EVs with different sizes in sample inlet, outlet A, outlet B, and that from ultracentrifugation, the results were calculated from the NTA measurements; B) TEM images of EVs in outlet B and that collected from ultracentrifugation; C) ExoView characterization of sEVs collected from outlet B and ultracentrifugation, scale bars represent 20 μm.

CONCLUSION

In this work, the separation system based on negative magnetophoresis achieves effective separation of sEVs from cell culture supernatant, meanwhile, the system ensures the viability of the sEVs. The separation of a variety of micro-/nano-scale biological samples using the proposed technique can be envisaged in future.

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CONTACT

* H. Yang; phone: +86-0755-86392675; hui.yang@siat.ac.cn